Catch Alpha: Testing the Evaluator Before Testing the Trading Strategy

An evaluation rule calibrated on synthetic nulls and planted edges before it is applied to market data.

Joris Viaud · Independent Researcher, France · ORCID 0009-0000-5757-269X · joris.viaud@edu.ece.fr

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Summary

Anyone who tries enough trading ideas will find one that looks great on past data, even if none of them works. We built a judge that counts every attempt and refuses to rule when the data are too short, and we tested the judge first: on simulated markets with no real edge it almost never accepted one, and it detected planted edges once they were large enough. It then judged four real strategies on twelve cryptocurrencies: the data show predictive patterns, but no tested portfolio established a reliable net profit after fees under the protocol.

Conclusions

1. Calibrate the decision rule on known nulls before touching market data.

2. Count every attempt; here the trial count is the safeguard that matters after selection.

3. Abstain when the sample is too short: ≥ 6.2 years of hourly data for a Sharpe of 1.

4. Predictability ≠ net profit: reversal and funding are real; the tested portfolios do not net a validated profit.

Motivation: search-inflated backtests

- **Search:** the best of 200 variants looks good even if none of them works.
- **Short samples:** annualizing scales noise as much as signal (×16 daily, ×94 hourly).
- **Late corrections:** correcting only the final strategy ignores earlier choices.

Origin: ~96 runs on 60 days of 15-min bars; Sharpe from −17 to +17.

Evaluation protocol: four checks and one diagnostic

1 Enough data?

Minimum length, one-sided mean test at $\alpha = 0.05$, power 0.80 (P = periods/yr, $S_{\min}$ = target annualized Sharpe):

$$n \;\geq\; 6.18\,\frac{P}{S_{\min}^2}$$

Hourly, $S_{\min} = 1 \Rightarrow$ **54,160 obs ≈ 6.2 years**. Our panel: 46,590 ⇒ INSUFFICIENT DATA: no Sharpe-based verdict.

2 Excess over benchmark

Stationary-bootstrap 95% confidence intervals (CIs) on mean and Sharpe of (candidate − benchmark): both lower bounds > 0. Blocks are resampled to preserve serial dependence; 400 replications (150 in the large calibration sweep).

3 Deflated Sharpe ratio (DSR)

Sharpe vs. the expected maximum of N zero-edge trials:

$$SR_0 = \sqrt{V}\left[(1-\gamma)\,\Phi^{-1}\!\left(1-\tfrac{1}{N}\right) + \gamma\,\Phi^{-1}\!\left(1-\tfrac{1}{Ne}\right)\right]$$

$$\text{pass if}\quad \Phi\!\left(\frac{SR - SR_0}{\sigma_{SR}}\right) \geq 0.95$$

$V \approx$ the candidate's own Sharpe variance; $\gamma \approx 0.5772$; σ_{SR} = standard error of the Sharpe.

***N* comes from an append-only trial ledger** counting every screen: real-data ledger: 353 recorded evaluations (352 selection evaluations + the single final-period run); 344 signal screens + 9 portfolio evaluations.

4 SPA (superior predictive ability) test

vs. every declared baseline, worst $p \leq 0.10$. Baselines declared before the run: cash, equal-weight buy-and-hold, and one strategy-specific comparator (e.g. the trend mirror for reversal, 168h momentum for raw carry).

5 PBO (probability of backtest overfitting)

Bailey et al. Build the matrix of per-fold scores (strategies × 6 folds); split the folds into every train/test half: $\binom{6}{3} = 20$ combinations. In each, take the strategy best on the train folds and record its relative rank ω on the test folds; $\ln\frac{\omega}{1-\omega} < 0$ means the in-sample winner fell below the out-of-sample median. PBO = share of the 20 combinations with a negative logit. Gate: PBO ≤ 0.50; a missing matrix blocks acceptance. Here: only two reversal variants have complete fold scores (2 × 6 matrix). **PBO = 0.00**; the in-sample winner's median out-of-sample rank is 0.67. Not statistically informative: it shows the diagnostic executed, not that selection bias is absent.

Procedure

- **Append-only trial ledger:** every signal screen and every gated evaluation is recorded; the ledger is the only source of N.
- Six chronological folds with 168-hour gaps at every boundary; a claim must repeat across folds.
- Final-period access budgeted to 3 looks; **1 used**, on frozen portfolios only.

Same Sharpe (0.06/period, 3,000 obs): DSR 0.99 if you count 3 trials, 0.63 if you count 347.

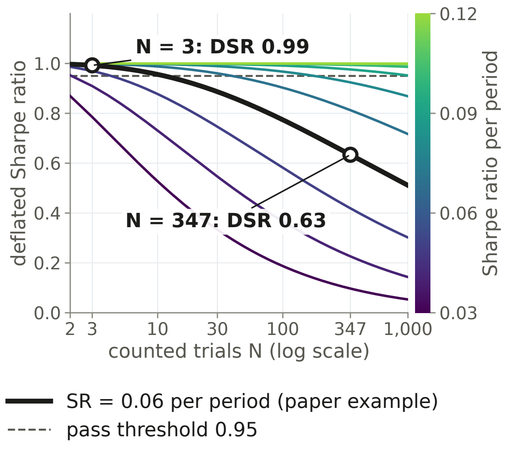


Fig. 1 – Same Sharpe, different trial count: DSR 0.99 at N = 3, 0.63 at N = 347 (347 = ledger count at the time of the paper's example; frozen count 353; 3,000 obs; computed from the paper's DSR formula).

References. Bailey & López de Prado 2014, deflated Sharpe ratio, *J. Portf. Manag.*; Bailey, Borwein, López de Prado & Zhu 2017, probability of backtest overfitting, *J. Comput. Finance*; Hansen 2005, superior predictive ability, *JBES*; Politis & Romano 1994, stationary bootstrap, *JASA*; Gârleanu & Pedersen 2013, dynamic trading with predictable returns and transaction costs, *J. Finance*.

Calibration on synthetic data

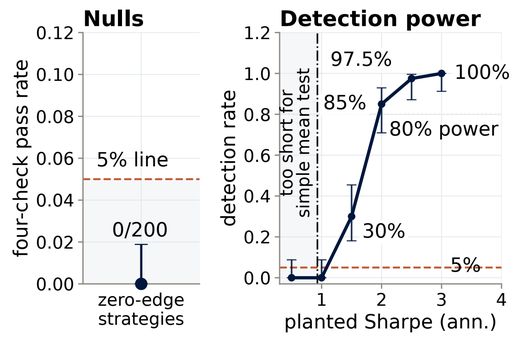


Fig. 2 – 0/200 zero-edge strategies accepted; detection rises with the planted edge. DSR counts N = 200 attempts; 40 draws per planted edge (paper Fig. 1).

- 200 zero-edge Gaussian strategies, 1,800 obs, N = 200: **0/200 accepted** (Wilson upper bound 1.9%). Calibrated on the tested generators only.
- Gaussian, Student-t_3, GARCH(1,1), AR(1) at 1,800 / 5,000 / 50,000 obs: **0/2,400** (1.9% upper bound per setting). Market-derived null (real signal shifted ≥ 26 weeks): 0/200.
- Planted annualized Sharpe 0.5 → 3.0, 40 draws each: detection **0 / 0 / 30 / 85 / 97.5 / 100%**.

Selection-aware calibration and ablation

3,300 campaigns; 200 equicorrelated candidates; winner = best development Sharpe. Rates in %; detection at planted S = 1.5.

	Null false accept.			Detection
Rule	ρ=0	0.3	0.7	ρ=0
On selection data				
Uncorrected t-test	100.0	83.6	48.2	86.5
Four checks, N=200	0.2	0.0	0.0	35.0
Four checks, N=1 (clamped to 2)	36.4	20.6	8.0	74.5
DSR alone, N=200	0.2	0.0	0.0	35.0
On an untouched holdout				
Four checks, N=200	0.0	0.0	0.0	24.5
t-test alone	5.0	5.6	7.6	86.0

The trial count provides the control: DSR alone ≈ four checks; drop N and **8–36%** of null winners pass. A clean single-use holdout is near nominal (5.0–7.6%) with more power: the correction protects selection on reused data; on a truly independent, single-use holdout a plain test is more powerful here.

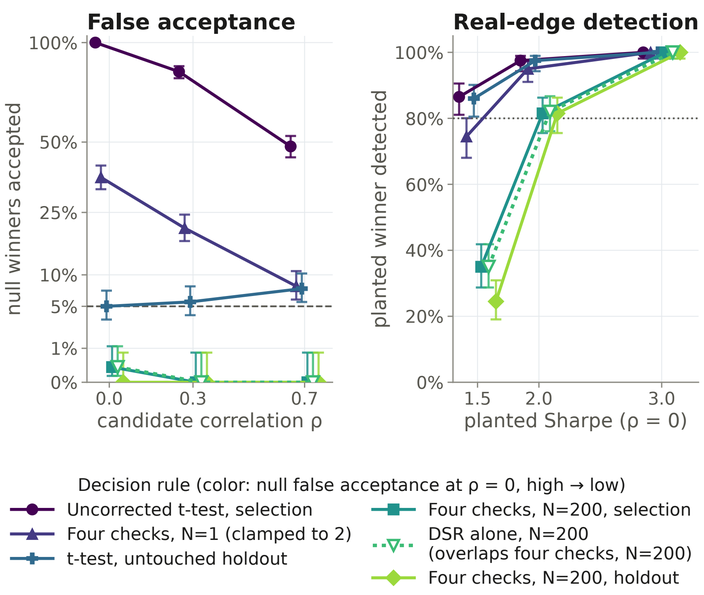


Fig. 3 – After selecting the best of 200 correlated candidates; 3,300 campaigns, Wilson 95% CIs, square-root vertical scale in the null panel (computed from the paper's committed data).

Scope of the evidence

- **Supported:** on the tested generators the four checks reject zero-edge winners (1 of 1,500 accepted after selecting the best of 200 correlated candidates); the trial count carries that control.
- **Not shown:** a false-positive bound for untested market processes, or that any adaptive search is covered. The simulated family is fixed.
- **Not shown:** a profitable strategy; the learning result is configuration-specific and the real-data result is conditional on this panel, horizon and 5 bps cost.

Learning test: offline RL on a planted signal

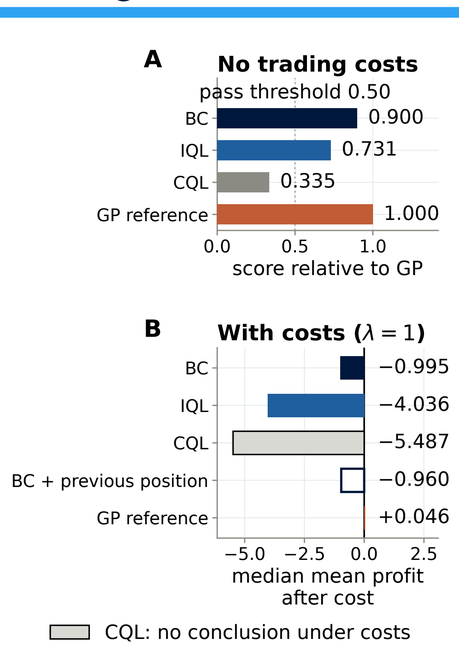


Fig. 4 – BC and IQL learn the signal but lose under quadratic costs in this configuration (paper Fig. 2).

5 assets, IC = 0.05 (information coefficient: corr(signal, next return)); every policy starts flat; 10 seeds × 100 paths.

- No costs, median Q (score ÷ GP score): BC 0.900, IQL 0.731, CQL 0.335 (threshold 0.50).
- λ = 1 (trading-cost weight): BC −0.995, IQL −4.036, **negative in 10/10 seeds**; turnover 6.03× / 11.97× that of GP; GP itself +0.046.
- BC with the previous position as input: −0.960.
- Learning the signal does not make it tradable after costs.

BC = behaviour cloning, IQL = implicit Q-learning, CQL = conservative Q-learning; GP = Gârleanu–Pedersen analytic reference.

Market (per asset): $s_{t+1} = \phi s_t + \eta_{t+1}, \qquad r_{t+1} = b\,s_t + \epsilon_{t+1}$

with $\phi = 0.9$, $\eta \sim N(0, 1-\phi^2)$, $b = 0.0005$, $\sigma_\epsilon = 0.01\sqrt{1-0.05^2}$, so IC = 0.05.

Score: $S(\pi) = \sum_t \left[w_t^\top r_{t+1} - \frac{\lambda}{2}\|w_t - w_{t-1}\|_2^2\right], \qquad Q(\pi) = \frac{S(\pi)}{S(\mathrm{GP})}$

Q is the relative score (threshold 0.50).

GP rule (partial adjustment): $w_t = w_{t-1} + A\,(\mathrm{aim}_t - w_{t-1}), \qquad \mathrm{aim}_t = m(\phi)\,\frac{b\,s_t}{g}$

with $g = \gamma_r \sigma_\epsilon^2$, risk aversion $\gamma_r = 10$. $\lambda = 0$: $A = 1$, $m = 1$ (pure Markowitz, aim = 0.50 s_t). $\lambda = 1$: $A = 0.031$, $m = 0.243$, aim = 0.12 s_t: **GP moves 3% of the way per period**; BC/IQL turnover is 6.03× / 11.97× that of GP.

GP optimises a risk-adjusted objective; S is raw profit net of costs, so Q is distance to a reference, not exact regret.

Real-data study: twelve Binance perpetuals

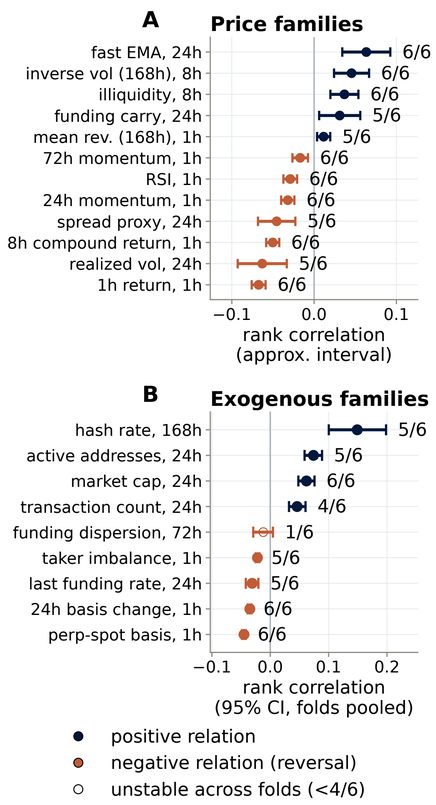


Fig. 5 – Rank correlation with future returns (paper Fig. 3). Price-panel bars are mean fold IC ± 1.96·|mean IC|/|median fold t|, an approximation rather than a 95% CI; external-panel bars combine fold-level Newey–West standard errors treating folds as independent; n/6 = folds whose Newey–West interval excludes zero; on-chain panel shown before market-cap removal.

- 128 price signal–horizon pairs; 69 fold-stable.
- 24h momentum → next hour: rank corr. **−0.032 in 6/6 folds** (median fold t = −7.79): short-horizon reversal.
- Tail portfolio (long losers, short winners): **−1.45 (24h) / −3.02 (8h)** before costs; the mirror trend book gains the same gross, so the tails trend at the tested horizons.
- 216 external pairs mostly reduce to size; after market-cap removal two small on-chain leads remain (hash rate 24h: 0.049; active addresses 72h: 0.061). These are reported as leads, not strategies.
- Final period (Jan 2025–Jun 2026, 12,648 h), one look: no interval above zero; the sample is too short.

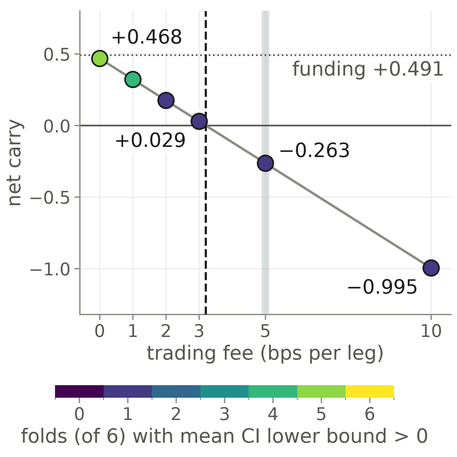


Fig. 6 – Cash-and-carry net return vs fee per leg; funding +0.491 is lost above about 3.2 bps/leg (computed from the paper's committed data).

- Funding positive in 6/6 folds; cash-and-carry receives +0.491, pays −0.023 basis and −0.732 fees: **nets −0.263**.
- Break-even ≈ **3.2 bps/leg** (frozen cost 5 bps).